Journal of Plastic, Reconstructive & Aesthetic Surgery 90 (2024) 227–239

ELSEVIER

JPRAS
An International Journal of
Surgical Reconstruction
www.JPRASurg.com

REVIEW

Merits of simulation-based education: A systematic review and meta-analysis

Jose Foppiani [a,1], Krystof Stanek [b,1], Angelica Hernandez Alvarez [a], Allan Weidman [a], Lauren Valentine [a], Irena J. Oh [c], Khaled Albakri [d], Umar Choudry [e], Carolyn R. Rogers-Vizena [b,f], Samuel J. Lin [a,f,*]

[a] *Department of Plastic and Reconstructive Surgery, Beth Israel Deaconess Medical Center, Boston, MA, USA*
[b] *Department of Plastic and Oral Surgery, Boston Children's Hospital, Boston, MA, USA*
[c] *Georgetown University School of Medicine, Washington, DC, USA*
[d] *Faculty of Medicine, The Hashemite University, Zarqa, Jordan*
[e] *Department of Plastic and Reconstructive Surgery, University of Minnesota, Minneapolis, MN, USA*
[f] *Department of Surgery, Harvard Medical School, Boston, MA, USA*

Received 6 May 2023; Accepted 29 January 2024

KEYWORDS
Surgical simulation;
Surgical education;
Meta-analysis;
OSATS

Summary *Background:* The drive to improve surgical proficiency through advanced simulation-based training has gained momentum. This meta-analysis systematically evaluated evidence regarding the impact of plastic surgery-related simulation on the performance of residents.
Methods: A systematic search of PubMed, Web of Science, and Cochrane Library and review of articles was performed following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis protocol. An inverse-variance random-effects model was used to combine study estimates to account for between-study variability. Objective structured assessment of technical skills (OSATS) scores and subjective confidence scores were used to assess the impact of the simulation with positive changes from the baseline indicating better outcomes.
Results: Eighteen studies pooling 367 trainees who participated in various simulations were included. Completion of simulation training was associated with significant improvement in subjective confidence scores with a mean increase of 1.44 units (95% CI: 0.93 to 1.94, $P < 0.001$), and in OSATS scores, with a mean increase of 1.24 units (95% CI: 0.87 to 1.62,

* Correspondence to: Department of Plastic and Reconstructive Surgery, Beth Israel Deaconess Medical Center, 110 Francis Street Suite 5A, Boston, MA, USA.
E-mail address: sjlin@bidmc.harvard.edu (S.J. Lin).
[1] J.F. and K.S. contributed equally to this work.

https://doi.org/10.1016/j.bjps.2024.01.021

$P < 0.001$), both on a 1-5 scale. Participants reported high satisfaction scores (mean = 4.76 units, 95% CI = 4.61 to 4.91, $P = 0.006$), also on a 1-5 scale.
Conclusions: Participation in surgical simulation markedly improved objective and subjective scoring metrics for surgical trainees. Several simulation devices are available for honing surgical skills, with the potential for advancements. The evidence demonstrates the effectiveness of simulations; thus, incorporating simulation into training curricula should be a priority in the field of plastic surgery.

Contents

The increasing need to adapt surgical education has arisen owing to the changing landscape brought about by work hour limitations and heightened administrative workload.[1-3] Teaching programs are progressively moving away from the conventional "see one, do one, teach one" method when instructing complex surgical skills and are instead embracing approaches that incorporate simulation-based training. For example, "The Fundamentals of Laparoscopic Surgery," developed by the Society of American Gastrointestinal and Endoscopic Surgeons is a well-established educational module that applies simulation to teaching and evaluating trainees.[4,5] Similarly, arthroscopy simulation in orthopedic surgery has become an obligatory part of certification by the Swiss Orthopaedics Board.[6,7] However, the abstract nature of plastic surgery procedures and the demand for highly realistic soft tissue models renders the implementation of simulation into formal training challenging.

Several different simulator types are currently used in plastic surgery training. Thomson et al. identified four basic categories: computer-based, animal, cadaver, and synthetic simulators; however, each category comes with its pros and cons.[8] Although virtual reality programs are most predominantly used, they are limited by the relatively high initial cost to the developer, and the lack of variability may conflate memorization with new skills acquisition. Animal models help cultivate skills that are easily transferable to the operating room (OR), but ethical concerns increasingly deter their implementation. Though cadavers provide the closest resemblance to live surgery, they are associated with limited availability, high financial costs to the user, and additional ethical requirements. Finally, synthetic simulators have evolved considerably over the past 5 years, especially in craniofacial surgery. Though emerging research on high-fidelity haptic simulators for cleft lip, cleft palate, and facial flaps has revealed that these are highly effective educational tools, their high cost impedes their wide-scale dissemination.[9-13] Regardless of these challenges, continued innovation is expanding opportunities for plastic surgery residents to train using simulation technology.

As simulation models are increasingly being incorporated into plastic surgery training, clarifying and quantifying their educational impact is necessary to inform when and how they are to be used. The present article primarily aimed to analyze the effect of simulation training on the performance and self-reported confidence of residents.

Methods

This study protocol was prospectively registered with PROSPERO (Study # ID: 42022363290) and was exempt from institutional review board consideration at our institution.[14] Methodology followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis statement guidelines.[14]

Eligibility criteria

Included studies were defined as those containing information on surgical trainees (residents and postgraduates) who participated in surgical skill simulation pertaining to the field of plastic surgery. This included simulations performed by trainees in other surgical fields such as vascular surgery and otolaryngology or head and neck surgery, as long as the simulation involved a skill or procedure that is also used by plastic surgeons in the United States (i.e., vascular anastomosis, tendon repair, and cleft palate repair). The full eligibility criteria are accessible at PROSPERO and are as follows:

Inclusion criteria:

- o Surgical trainees (residents and postgraduates) and attendees
- o All types of surgical simulation, virtual or augmented reality and applications that involve a procedure or skill that is required or performed by plastic surgery residents
- o Observational studies and clinical trials
- o Studies in English, French, and Spanish

Exclusion criteria:

- o Editorials, commentary reports, abstracts, and letters to the editor
- o Medical students

Search strategy

A comprehensive research review using subject headings, controlled vocabulary, and keywords was conducted on September 25, 2022, on MEDLINE (in Ovid), Web of Science, and the Cochrane Central Register for studies published until September 2022. Our full-text search strategy is accessible at PROSPERO.

Study selection

The search results were uploaded into the online systematic review program Covidence to conduct study selection.[14] Three independent reviewers (K.S., K.A., and J.A.F.) performed a two-step screening process for study selection. First, titles and abstracts were screened. If conflicts arose between the two reviewers, the third reviewer moderated a discussion to resolve the conflict via joint decision. Next a full-text analysis was performed by the three reviewers (J.A.F, A.H.A., and K.S.). If conflicts arose between the reviewers, the third reviewer moderated a discussion to reach a joint decision.

Data extraction or synthesis

Data extraction was guided by a predetermined checklist: last name of the first author, year of publication, type of simulation or device used, level of resident seniority, total number of participants, gender, age, number of simulations per participants, duration of the simulation, simulation protocol, outcomes of the simulations (objective assessment, subjective assessments, and relevant qualitative outcomes).

Outcomes

The primary outcomes of this systematic review were objective and subjective changes in performance of residents following surgical simulation training. Incorporated articles employed a diverse array of objective assessment measures, which used different scoring systems rather than a singular metric. To address this heterogeneity, we conducted a subgroup analysis of the studies by employing comparable tools and scales to ensure a more nuanced and precise interpretation of the data. They included self-reported changes in surgical skill confidence and change in overall average objective structured assessment of technical skills (OSATS) scores. Secondary outcomes were participant satisfaction with the simulation and cost of the simulation.

Quality assessment

To assess the risk of bias, the National Institute of Health (NIH) quality assessment tool was used.[19] Each article was categorized as follows: "low risk," "moderate risk," or "high risk" of bias.

Statistical analysis

An inverse-variance random-effects model was used to combine study estimates. This statistical model was chosen because it accounts for the between-study variability, assuming that the treatment effect may vary from investigation to investigation.[15] We used the restricted maximum-likelihood method to estimate between-study variance, as this approach provides less biased estimates.[16]

For subjective confidence and OSATS scores, impact of simulation training was measured using the mean difference from the baseline (within-group change from baseline) with positive values indicating better outcomes. All scores were converted to a 1–5 scale to standardize the data and perform all analyses. The mean change was approximated based on baseline and follow-up scores, assuming a correlation coefficient of 0.5.

For post-training satisfaction, follow-up scores were used. In cases with missing standard deviations (SDs), sample size weighted average of SDs from studies that provided complete information were used to impute missing SDs (time-point specific: baseline and follow-up). The summary results were presented as mean changes from baseline (95% confidence interval, 95% CI) or post-training means (95% CI).

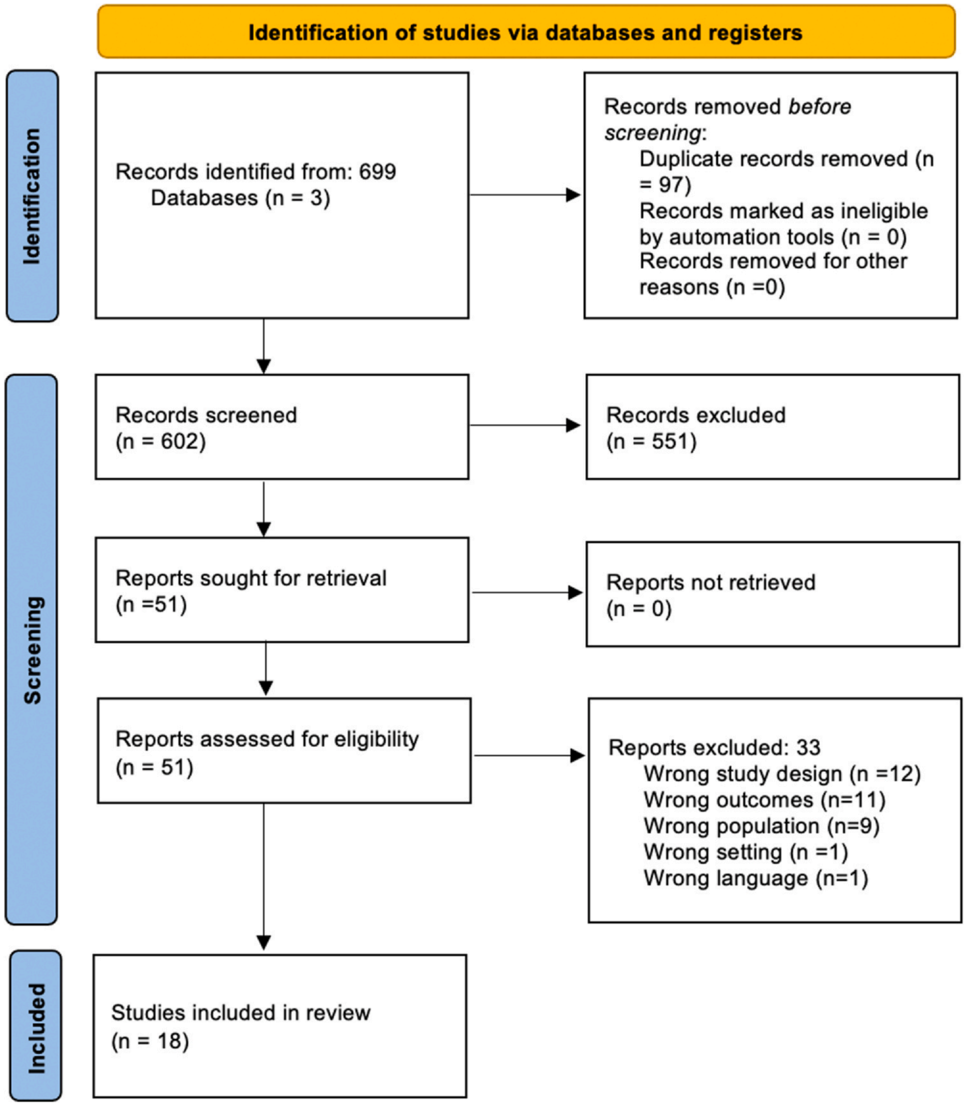

Figure 1 PRISMA flow diagram.

The presence of statistical heterogeneity among study estimates was determined via the Cochran's Q test, whereas the between-study variance was quantified using the I^2 statistic.[17,18] For the Q test, a *P*-value < 0.10 was considered a statistically significant heterogeneity.[17] In addition, 95% predictive intervals (95% PI) were used, which describe a range with 95% probability of containing the future treatment effects to be estimated in a new study or setting.[19] We did not investigate funnel plot asymmetry owing to the limited number of studies available (i.e., < 10 estimates).[20] For summary estimates, two-sided *P*-values < 0.05 were considered statistically significant. All analyses were performed using Stata (version 16, College Station, TX, USA).

Results

This systematic review initially identified a total of 699 publications through a comprehensive literature search. After removing 97 duplicate records, 602 articles were subjected to further screening. Among these, 551 publications were excluded as they did not meet the inclusion criteria and were deemed irrelevant to the research objectives. The remaining 51 articles underwent a full-text screening, where an additional 33 publications were removed owing to the failure to meet the established inclusion criteria. The systematic review ultimately included 18 high-quality, relevant publications (Figures 1-4).[21-38]

Qualitative assessment

The systematic review consisted of data from 367 surgical trainees who participated in a total of 521 simulations. Most participants were plastic surgery residents (n = 189) at different stages in their training or postgraduate year (PGY). The remaining 178 participants included trainees from various surgical subspecialties (such as general surgery, ENT, orthopedic surgery, neurosurgery, cardiac surgery, urology, and obstetrics and gynecology) and attending physicians. Table 1 summarizes the recorded data on the participants' sex and age, which were only reported in five and four

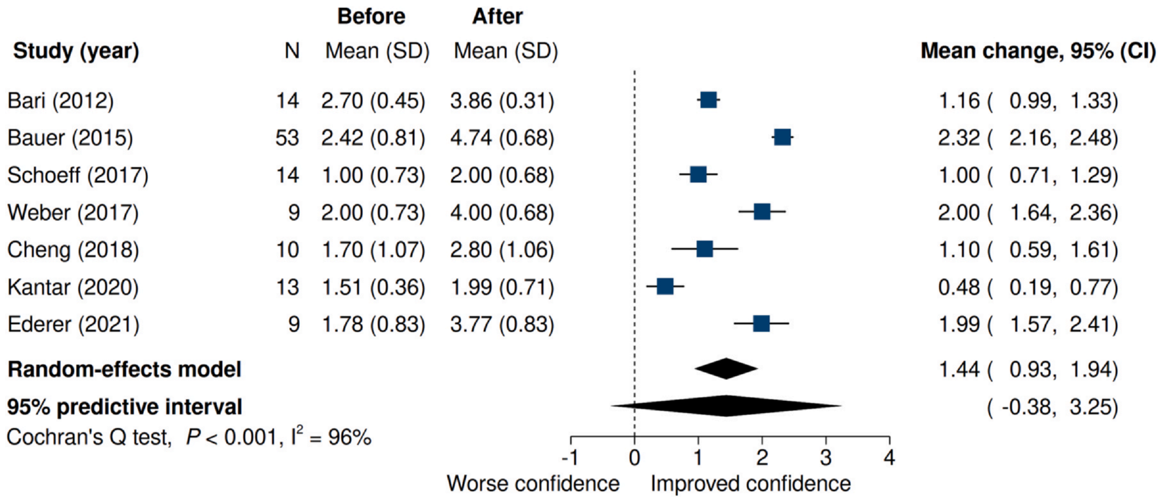

Figure 2 Forest plot showing the mean change from baseline in subjective confidence scores. The terms “Before” and “After” relate to the subjective scores obtained before and after completing a simulation training program, respectively. The 95% CI denotes 95% confidence interval. SD denotes standard deviation.

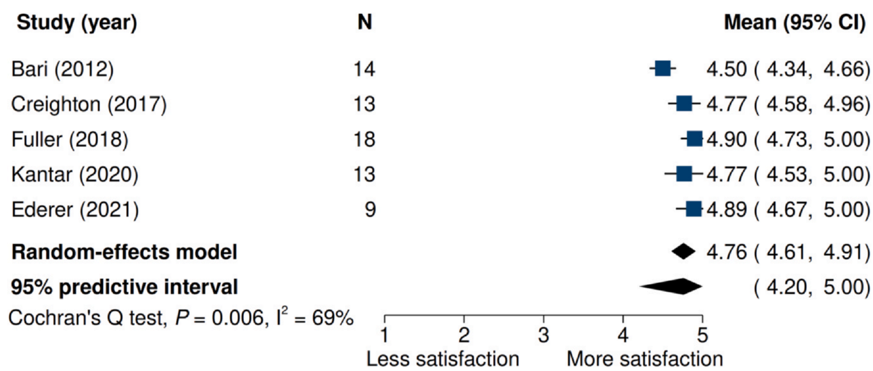

Figure 3 Post-training satisfaction (subjective outcome). The 95% CI denotes 95% confidence interval. SD denotes standard deviation.

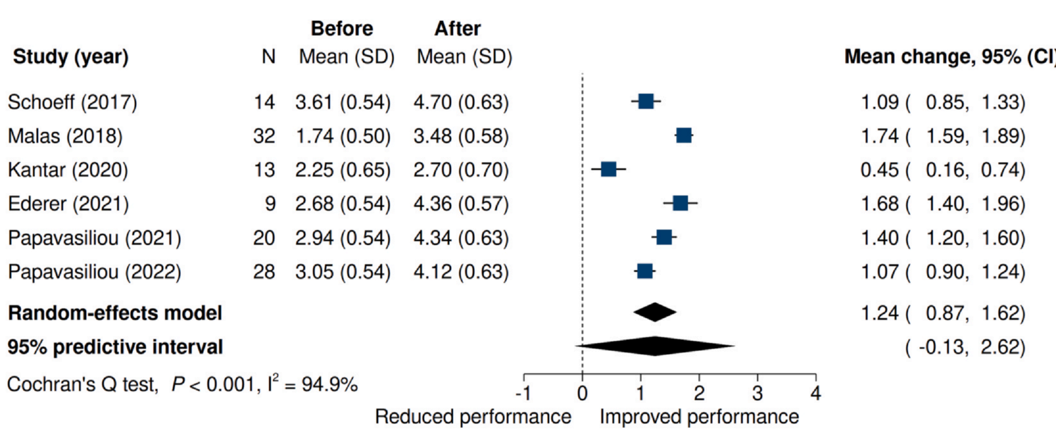

Figure 4 Forest plot showing the mean change from baseline in objective structured assessment of technical skills (OSATS) scores. The terms “Before” and “After” refer to the OSATS scores obtained before and after completing a simulation training program, respectively. The 95% CI denotes 95% confidence interval. SD denotes standard deviation.

studies, respectively. Additionally, the table includes the results of the NIH study quality assessment.

A total of nine studies included in this systematic review used synthetic models whereas the other nine used biological models. All simulations were physical-based simulations, and none used virtual or augmented reality simulations. The studies were conducted over diverse range of time periods, with a minimum duration of 1 day to a

Table 1 Study demographics.

Authors, years	Simulator	Plastic surgery residents (n = 186)	Residents from other surgical specialities (n = 178)	Attendings (n = 26)	Level of seniority (PGY)	Number of simulations (n = 521)	NIH quality assessment
Price, 2011	Live porcine model	1	38		PGY 1-2	39	High
Malas, 2018	Synthetic vessel model	2	30		PGY 1-2	64	Low
Podolsky, 2018	Synthetic cleft palate model	6		2	PGY 3-7	28	Low
Ederer, 2021	Human skin	9			PGY 1-6	18	High
Weber, 2017	Cadaver	9			PGY 1-7	26	Moderate
Cheng, 2018	Synthetic cleft palate model	10			PGY 1-7	10	Moderate
Odobescu, 2019	Thiel cadaveric peripheral nerve model	11	5		PGY 1-6	16	Moderate
Kantar, 2020	3D haptic cleft palate model	13			PGY 1-4	13	Low
Bari, 2012	Fresh cadaver	14			PGY 1-6	36	Low
Saldanha, 2021	Synthetic cleft lip model	16			PGY 3-7	16	Moderate
Papavasiliou, 2021	Synthetic flexor tendon model	20			PGY 1-6	20	Low
Papavasiliou, 2022	Synthetic hand model	28			SHO 1-2	28	Moderate
Wang, 2017	Mannequins with t-shirts	50			PGY 1	50	High
Bauer, 2015	Cadaver		31	22		8	Moderate
Creighton, 2017	Chicken thigh model		13		PGY 2-5	39	Moderate
Fuller, 2018	Synthetic rib cartilage and nasal simulator		18		PGY 1-2	36	Low
Pafitanis, 2019	Chicken thigh model and silicone vessel		3	2		60	High
Schoeff, 2017	Chicken thigh model		14		PGY 1-5	14	Moderate

Table 2 Simulation protocols.

Authors, years	Device	Duration of training/ simulation	Training protocol
Weber, 2017	Human cadaver	4 weeks	Participants trained on various procedures on cadavers.
Wang, 2017	Mannequins	15 days	Participants attended a 30-minute lecture, then performed breast augmentation on fiberglass mannequins using T-shirts and elastic compression garments.
Schoeff, 2017	Chicken thigh	Three sessions of 30 min, 1 week apart each.	Participants were randomized in an “Interval” or “Massed” group. The interval group performed three separate 30-minute microvascular anastomosis practice sessions and the massed group performed a single 90-minute practice session.
Saldanha, 2021	High-fidelity cleft lip simulator	3 h-sessions every 2–3 months	Participants received an introductory 45-minute lecture and then performed unilateral cleft lip repair on the simulator.
Price, 2011	Live porcine model	2 weeks	Participants performed an end-to-side microvascular anastomosis in an in vivo model. One group received an expert guided tutorial alone whereas another was allowed to self-train on the simulation models.
Podolsky, 2018	High-fidelity cleft palate simulator	10 weeks	All residents performed five simulated cleft palate repairs, with each session 2 weeks apart.
Papavasiliou, 2022	3D-printed flexor tendon model	3 h	A 3-hour course was run remotely via Zoom. The model was used for flexor tendon repair in 4 levels of difficulty.
Papavasiliou, 2021	3D-printed hand model	Not reported	Participants performed K-wire fixation of a hand fracture on a 3D printed model in 6 levels of difficulty.
Pafitanis, 2019	Three-layered silicone vessel and chicken thigh vessel	1 month	Participants performed consecutive anastomoses using the coupler device following expert instruction until they reached the experts’ hand motion analysis performance.
Odobescu, 2019	Thiel cadaveric peripheral nerve model	Not specified	Participants completed an end-to-end nerve coaptation, which was recorded and then evaluated by 5 fellowship-trained microsurgeons.
Bari, 2012	Human cadaver	1 day (14 subjects) + 1 day 6 months later (8 subjects)	Fourteen participants performed 1 pre-tutorial tendon repair followed by 1 post-tendon repair. Eight residents from the same cohort performed 1 more tendon repair 6 months after the tutorial.
Bauer, 2015	Human cadaver	3 day training	Subjects were shown how to raise flaps in steps, and then they performed those steps themselves under supervision.
Cheng, 2018	Cleft palate simulator	1 day	Following a demonstration of the 15 critical steps of palatoplasty, each participant performed the procedure on the simulator.
Creighton, 2017	Chicken thigh model	1 day	A previously validated chicken thigh model was used to simulate microvascular anastomosis with the TCC of the first anastomosis compared with the TCC of their last anastomosis

(*continued on next page*)

Table 2 (*continued*)

Authors, years	Device	Duration of training/ simulation	Training protocol
Ederer, 2021	Human skin	23 months	Excised human skin from patients who underwent post-bariatric body contouring surgery were utilized for the simulation which consisted of theoretical local flap performance.
Fuller, 2018	Synthetic rib cartilage for ear carving, septoplasty simulator,	3 days	Multidays course with various simulations, lectures, and assistance in surgery.
Kantar, 2020	3D Haptic model	1 day	Education with either software or textbook followed by surgical simulation on a 3D cleft device.
Malas, 2018	Synthetic vessel model (polytetrafluoroethylene grafts)	1 week	Participants received a didactic training session, and then practiced on bench models, with one group receiving instructional media supplementation and the other acting as control.

TCC, timing to completion.

maximum of 23 months. A total of 15 studies included some form of education either via expert instruction, textbook, or instructional videos (Table 2).

All 18 studies included either an objective or subjective evaluation of the participants' performance post-simulation. Self-reported confidence in performing a skill or procedure pre- and post-simulations was reported in seven studies. Satisfaction with the simulations was reported in five studies. Lastly, six studies reported pre- and post-simulation OSATS scores (Table 3).

Quantitative assessment

Completion of a simulation program was associated with a statistically significant improvement in subjective confidence scores with a summary mean change from baseline to 1.44 units (95% CI: 0.93 to 1.94, $P < 0.001$) on a 1-5 scale. However, there was evidence that improvements varied more than expected by chance alone (Cochran's Q test, $P < 0.001$ and $I^2 = 96\%$), with considerable between-study heterogeneity. Given the available evidence, the 95% PI ranged from −0.38 (negative improvement in confidence) to 3.25 (positive improvement in confidence). Based on this evidence, there is a 98% chance that in future studies the true underlying improvement in subjective confidence scores would be positive (i.e., an increase larger than zero).

Completion of a simulated program was associated with high satisfaction scores (mean = 4.76 units, 95% CI = 4.61 to 4.91) on a subjective 1-5 scale. Mean satisfaction scores showed high statistical heterogeneity between studies, as indicated by a significant Cochran's Q test ($P = 0.006$) and a large I^2 (69%). However, this statistical variability is unlikely to have a significant impact on the instructional process (i.e., it is didactically not relevant). Based on the available evidence, there is 95% confidence that post-training satisfaction scores in future studies would fall within a range of 4.2 to 5.0.

The completion of a simulation program was also associated with a statistically significant improvement in OSATS scores, with an average increase of 1.24 units (95% CI: 0.87 to 1.62, $P < 0.001$) on a 1-5 scale. Study estimates varied more than expected by chance (Cochran's Q test, $P < 0.001$ and $I^2 = 94.9\%$), indicating a high level of heterogeneity between studies. The 95% PI, which represents the range of likely improvement in performance in future studies and settings, ranged from −0.13 (reduced to near null improvement) to 2.62 (significant improvement). Based on the available studies, there is a 99.4% probability that the mean change from baseline in OSATS scores in future studies would be greater than 0.

Discussion

The motivation to improve surgical proficiency through advanced simulation-based training has gained significant momentum in recent years. This momentum is reflected in the development of diverse simulation devices to enhance various surgical skills, with promise for more advancements in the future. This systematic review found that participating in surgical simulation resulted in marked improvement in objective and subjective scoring metrics among surgical trainees. Simulations not only improved skills and confidence, but the participants also reported that the experience was highly enjoyable. Given these positive outcomes, plastic surgery programs can take advantage of the available simulation tools by systematically incorporating them into their training curriculum.

OSATS score is a commonly used tool for evaluating surgical skills. It is a validated, objective, and structured assessment that entails observing and scoring the performance of candidates during simulated surgical tasks. This review found an average increase in OSATS score by 1.24/5 units ($P < 0.001$) among the studies that reported OSATS scores. This substantial improvement in objective scoring metric reflects the capacity of simulations to enhance surgical skills in residents. Plastic surgery is a highly technical specialty with a steep learning curve, particularly when it comes to mastering microsurgical skills, which require extensive practice over several years. By providing residents with the opportunity to hone these skills using

Table 3 Summary of results.

Authors, years	Subjective assessment	Objective assessment - results
Weber, 2017	Surveys showed increased confidence after simulation (2 to 4, $P < 0.01$), with residents favoring cadaveric simulation and reporting improved technique, speed, safety, and anatomy knowledge.	NA
Wang, 2017	There was a significant difference ($P < 0.05$) before and after the training regarding candidate confidence, anatomical awareness, and endoscope control including the dexterity and hand-to-eye coordination	NA
Schoeff, 2017	Subjective confidence in micro instrument handling, microvascular anastomosis, and operative microscopy increased significantly after training ($P < 0.001$), with no differences between groups.	The entire cohort showed significant average increases in TSS and GRS. The interval group improved significantly, while massed group's improvement was not statistically significant. Seniors outperformed juniors in both metrics.
Saldanha, 2021	Simulation significantly improved knowledge or skill in 5 of 6 domains. Higher training levels correlated with higher self-assessment scores ($r = 0.62$, $P = 0.011$) and self-confidence ($r = 0.31$, $P = 0.24$).	Half (50%) of trainees “successfully” attained lip length and nostril shape measurements within the range of the 3 attending surgeons, whereas just under half (41%) performed within the attending range for nasal deviation
Price, 2011	NA	Residents with expert-guided and independent simulator training scored higher on OSATS and end-product evaluations than those with expert-guided training alone. They also performed anastomosis significantly faster.
Podolsky, 2018	NA	Residents' procedure-specific, global rating, and end-product scores increased logarithmically after each simulation session. Six to three sessions are required to reach the minimum standard, using established cutoff scores.
Papavasiliou, 2022	NA	The overall average knowledge-based scores of the cohort pre- and post-assessment were 1.48/5 (29.6%) and 3.56/5 (71.5%), respectively. The overall average skills-based scores of the cohort pre- and post-assessments were 3.05/5 (61%) and 4.12/5 (82.5%), respectively. Significant ($P < 0.01$) difference of improvement of knowledge and skills was noted on all trainees.
Papavasiliou, 2021	NA	The overall average scores of the cohort before and after assessment were 23.75/40 (59.4%) and 34.7/40 (86.8%), respectively. Significant ($P < 0.01$) difference of improvement of skills was noted on all trainees
Pafitanis, 2019	NA	Novices needed an average of 8 and 4 consecutive repetitions in phase 1 and 2 to reach expert performance, with improvements of 69% and 37%, respectively. Adequate flow measures were achieved in phase 3, with 12.5% of coupler applications successful during the exit performance-phase 3, and 7 out of 8 demonstrating an adequate range of flow measures during the end-product assessment with f-cMP.
Odobescu, 2019	NA	The average MNES was significantly higher in senior residents than in junior residents (exact $P = 0.0034$, Wilcoxon test). Average MNES was also significantly associated with the self-reported level of experience of the resident (exact P ¼.0034, Kruskal-Wallis test)

(*continued on next page*)

Table 3 (*continued*)

Authors, years	Subjective assessment	Objective assessment - results
Bari, 2012	Overall, residents were satisfied to very satisfied with the educational module (4.5 ± 0.3). Self-rated confidence in performing a zone II flexor tendon repair improved from 2.7 ± 0.45 to 3.86 ± 0.31 on a five-point Likert scale ($P < 0.01$).	Force needed for a 2-mm gap and ultimate breaking strength improved significantly after the tutorial. Pre-tutorial, senior residents had stronger repairs than junior residents, but post-tutorial, differences diminished, showing training level had little impact on post-tutorial strength.
Bauer, 2015	There was a significant increase in perceived ability for all free flap placements except the lateral arm flap ($P = 0.052$). Self-assessment differed significantly between residents and consultants in radial forearm and iliac crest free flaps before the course ($P = 0.006$), but not afterward.	N/A
Cheng, 2018	Post-session, all participants reported significantly higher procedural confidence ($P < 0.05$). Trainees with prior cleft surgery experience had higher baseline confidence and post-session confidence.	90% of participants scored higher on the knowledge test after the session, with a 15% mean increase ($P = 0.05$). Trainees with prior cleft surgery experience showed a greater increase in knowledge and higher final test scores compared to those without experience, although their baseline knowledge was not significantly different ($P > 0.05$).
Creighton, 2017	All participants agreed that they improved at microvascular anastomosis during simulation, found it effective for learning surgical skills, and considered the model tissue realistic.	Participants' average time-to-complete (TTC) improved by 48.7%, with similar improvements for junior and senior residents. Both groups' first and final stitch TTCs did not significantly differ, showing comparable learning progress for all residents.
Ederer, 2021	Fresh human skin training model increased confidence in performing surgery, was more effective than alternatives, and was recommended for residency programs by participants.	Surgical skills evaluation showed a significant increase in mean scores for all criteria after training. Pre-training, residents struggled with accuracy and wound coverage, whereas post-training, their scores indicated good to very good procedural skills, with a significant increase in modified OSATS.
Fuller, 2018	The resident feedback was exceedingly positive with average scores for each component of the teaching module ranging from 8.9 to 9.8 on a Likert scale from 0 (very poor) to 10 (excellent), with the highest scores being given to the practical simulation workshops.	All residents completed pre- and post-module tests, with significant improvements across nearly all testing modalities. A significant difference in premodule written microtia test scores was observed between first- and second-year residents, but not in other scores.
Kantar, 2020	Participants found digital simulation more stimulating, engaging, clear, and effective for learning surgical skills than textbooks, and were more likely to recommend it.	Post-intervention improvements were seen in surgical knowledge, procedural confidence, and markings performance for all participants. The digital simulation group showed significant improvements in all areas, whereas the textbook group did not. Strong interrater reliability was observed.
Malas, 2018	NA	Pretest scores were similar for both groups, but post-intervention showed significant improvement in OSATS and EPRS scores for both. Visualization multimedia led to greater increases in these scores, whereas time to completion differences were nonsignificant.

simulators early in their careers, they can gain a head start and accumulate more hands-on experience. Interestingly, Niitsu et al. conducted a study to analyze changes in OSATS scores among 10 of their residents after serving as first assistants in 888 cases over a 3-year period (PGY3-PGY5).[39] They found a statistically significant change from 2.43/5.00 to 3.43/5.00 in their OSATS score ($P = < 0.001$). Notably, the change in OSATS score by 1.00 unit over 3 years reflects the evaluation of their skills across numerous procedures and techniques. Moreover, it indicates that simulation can result in improvement in technical skills that are comparable to the traditional hands-on skill acquired in the OR.

Our pooled analysis demonstrated that surgical simulation can improve OSATS scores such that they are comparable to

Table 4 Strengths of simulation as reported in the literature.

Strengths of surgical simulation	
Safety	Safe and controlled environment without the risk of harm to patients. This allows trainees to make mistakes and learn from them without causing any harm.
Efficiency	Enables repetition of procedures multiple times in succession as opposed to waiting for an opportunity to practice them in the OR
Improved skill acquisition	Trainees can gain hands-on experience with various surgical instruments and techniques, which can improve their overall surgical skill.
Assessment	Simulations can provide feedback on areas for improvement. This allows trainees to understand their strengths and weaknesses and focus on areas that need more attention.
Cost-effective	Simulations can be less expensive than live surgeries, as they do not require real patients or anesthesia.

those obtained via conventional surgical education. This finding supports surgical simulation as an efficient teaching tool for enhancing the surgical skills of trainees. Moreover, it underscores the potential of simulations to augment traditional learning experience in a safe, controlled environment for skill development. Ultimately, this congruence in outcomes highlights the need to integrate surgical simulation into surgical education programs to better prepare trainees for real-life OR procedures.

Furthermore, the potential impact of simulation on surgical skills was quantified using the 95% PI. The PI for OSATS score indicates a 99.4% probability of significant improvement by up to 2.62 units (out of 5) from each resident's baseline following simulations training. Therefore, surgical simulation can be confidently incorporated into the educational policies with minor risk and high potential for significant gain. Surgical education policy encompasses the standards for surgical training programs, criteria for awarding surgical qualifications, and procedures for accrediting surgical training institutions. The purpose of surgical education policy is to ensure that surgeons receive comprehensive and high-quality education that prepares them to provide safe and effective surgical care to patients. It is crucial that such policies evolve over time and incorporate changes that ultimately benefit the patient population. Given the marked objective improvement in the performance of residents following surgical simulation, and the low risk associated with its introduction into educational curricula, the Accreditation Council of Graduate Medical Education and surgical education societies should emphasize the utilization of simulation-based training in their educational programs.

One commonly explored, but undervalued benefit of surgical simulation is its impact on self-confidence (Table 4).[40] Indeed, evidence from the pooled literature demonstrates that completion of a simulation program is associated with significant improvement in self-perceived confidence scores with a mean change from baseline to 1.44 units (95% CI: 0.93 to 1.94, $P < 0.001$) on a 1-5 scale. Simulation training offers a unique opportunity for deliberate practice in a low-stakes environment, and this is critically important because practice and preparation have been shown to be cornerstones for building confidence.[41] Similarly, Elfenbein et al. demonstrated that the lack of confidence and readiness to practice independently can influence graduating surgeons to pursue subspecialty training.[42] Thus, participation in simulation programs offers an additional opportunity for budding surgeons to individually gain confidence that can carry over into their careers as independently practicing physicians.

Limitations

The results of this systematic review and meta-analysis should be viewed within the context of the study's limitations. Given the outcome being investigated and paucity of studies in this field, it was not feasible to restrict the systematic review to only randomized controlled trials or case-control studies, which led to high inter-study heterogeneity. The resulting population of participants was thus obtained largely from observational studies, which present biases inherent to their design and often have incomplete data. Although some studies used OSATS scoring, several studies only reported subjective assessments, and therefore, they could not be included in the statistical analysis. Consequently, this review was could not analyze sub-groups of simulation tools and their specific effects. This review accounted for the heterogeneity among the included studies by using random-effects models, which provided a robust method to ensure the validity of the results. Unfortunately, most studies did not report on the cost of the simulations. Thus, more surgical simulation studies using similar methodology are needed to evaluate individual subtypes of simulation, best models for simulation (synthetic vs. biological), and cost efficiency.

Conclusions

Surgical simulations that target skills relevant to plastic and reconstructive surgery offer marked improvements in objective and subjective scoring metrics for surgical trainees. A variety of simulation devices are available for honing diverse surgical skills, with high potential for more advanced models. To maximize the benefits of the teaching methods, incorporating simulation tools into training curricula should be a priority in surgical residency programs. The extensive range of available didactic and independent simulations demonstrates that they can be effectively integrated into the curricula while also serving as resources that facilitate self-study for residents to address their areas of weakness and enhance their learning experience.

Ethical approval

Not required.

Declarations of interest

None.

Funding

None.

Acknowledgments

We sincerely thank Victor Joao, affiliated with the Universidad De La Republica Uruguay, for his help as the active librarian for this project. He helped with the creation of search strategies and for gathering the search results for this manuscript. We would also like to thank Tiago Pereira for his expertise as the biostatistician of this project. We have no funding to disclose.

Author contributions

J.F. and K.S. designed the study, optimized the literature search, acted as reviewers, and resolved conflicts regarding the inclusion of papers. I.J.O. and K.A. worked on abstract screening. A.H.A., A.W., and L.V. worked on the manuscript. U.C., S.J.L., and C.R.R.V. oversaw all aspects of study design and SR writing. All authors have read and approved the final manuscript.

References

1. Dozois EJ, Holubar SD, Tsikitis VL, et al. Perceived impact of the 80-hour workweek: five years later. *J Surg Res* 2009;**156**:3–15.
2. Hutter MM, Kellogg KC, Ferguson CM, Abbott WM, Warshaw AL. The impact of the 80-hour resident workweek on surgical residents and attending surgeons. *Ann Surg* 2006;**243**:864–71. discussion 71-5.
3. Jamal MH, Wong S, Whalen TV. Effects of the reduction of surgical residents' work hours and implications for surgical residency programs: a narrative review. *BMC Med Educ* 2014;**14**(Suppl 1):S14.
4. Zendejas B, Ruparel RK, Cook DA. Validity evidence for the fundamentals of laparoscopic surgery (FLS) program as an assessment tool: a systematic review. *Surg Endosc* 2016;**30**:512–20.
5. Hur HC, Arden D, Dodge LE, Zheng B, Ricciotti HA. Fundamentals of laparoscopic surgery: a surgical skills assessment tool in gynecology. *JSLS* 2011;**15**:21–6.
6. Gallagher K, Bahadori S, Antonis J, Immins T, Wainwright TW, Middleton R. Validation of the hip arthroscopy module of the VirtaMed virtual reality arthroscopy trainer. *Surg Technol Int* 2019;**34**:430–6.
7. VirtaMed. https://www.virtamed.com/en/. Accessed 09/24/2022, 2022.
8. Thomson JE, Poudrier G, Stranix JT, Motosko CC, Hazen A. Current status of simulation training in plastic surgery residency programs: a review. *Arch Plast Surg* 2018;**45**:395–402.
9. Podolsky DJ, Wong Riff KW, Drake JM, Forrest CR, Fisher DM. A high fidelity cleft lip simulator. *Plast Reconstr Surg Glob Open* 2018;**6**:e1871.
10. Rogers-Vizena CR, Saldanha FYL, Hosmer AL, Weinstock PH. A new paradigm in cleft lip procedural excellence: creation and preliminary digital validation of a lifelike simulator. *Plast Reconstr Surg* 2018;**142**:1300–4.
11. Podolsky DJ, Fisher DM, Wong KW, Looi T, Drake JM, Forrest CR. Evaluation and implementation of a high-fidelity cleft palate simulator. *Plast Reconstr Surg* 2017;**139**:85e–96e.
12. Mohd Slim MA, Hurley R, Lechner M, Milner TD, Okhovat S. A systematic review of facial plastic surgery simulation training models. *J Laryngol Otol* 2022;**136**:197–207.
13. Powell AR, Srinivasan S, Green G, Kim J, Zopf DA. Computer-aided design, 3-D-printed manufacturing, and expert validation of a high-fidelity facial flap surgical simulator. *JAMA Facial Plast Surg* 2019;**21**:327–31.
14. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71.
15. Borenstein M, Hedges LV, Higgins JP, Rothstein HR. A basic introduction to fixed-effect and random-effects models for meta-analysis. *Res Synth Methods* 2010;**1**:97–111.
16. Langan D, Higgins JPT, Jackson D, Bowden J, Veroniki AA, Kontopantelis E, et al. A comparison of heterogeneity variance estimators in simulated random-effects meta-analyses. *Res Synth Methods* 2019;**10**:83–98.
17. Pereira TV, Patsopoulos NA, Salanti G, Ioannidis JP. Critical interpretation of Cochran's Q test depends on power and prior assumptions about heterogeneity. *Res Synth Methods* 2010;**1**:149–61.
18. Huedo-Medina TB, Sánchez-Meca J, Marín-Martínez F, Botella J. Assessing heterogeneity in meta-analysis: Q statistic or I^2 index? *Psychol Methods* 2006;**11**:193–206.
19. IntHout J, Ioannidis JPA, Rovers MM, Goeman JJ. Plea for routinely presenting prediction intervals in meta-analysis. *BMJ Open* 2016;**6**:e010247.
20. Sterne JA, Egger M, Smith GD. Systematic reviews in health care: investigating and dealing with publication and other biases in meta-analysis. *BMJ* 2001;**323**:101–5.
21. Price J, Naik V, Boodhwani M, Brandys T, Hendry P, Lam BK. A randomized evaluation of simulation training on performance of vascular anastomosis on a high-fidelity in vivo model: the role of deliberate practice. *J Thorac Cardiovasc Surg* 2011;**142**:496–503.
22. Malas T, Al-Atassi T, Brandys T, Naik V, Lapierre H, Lam BK. Impact of visualization on simulation training for vascular anastomosis. *J Thorac Cardiovasc Surg* 2018;**155**:1686–1693.e5.
23. Podolsky DJ, Fisher DM, Wong Riff KW, et al. Assessing technical performance and determining the learning curve in cleft palate surgery using a high-fidelity cleft palate simulator. *Plast Reconstr Surg* 2018;**141**:1485–500.
24. Ederer IA, Reutzsch FL, Schäfer RC, et al. A training model for local flaps using fresh human skin excised during body contouring procedures. *J Surg Res* 2021;**262**:190–6.
25. Weber EL, Leland HA, Azadgoli B, Minneti M, Carey JN. Preoperative surgical rehearsal using cadaveric fresh tissue surgical simulation increases resident operative confidence. *Ann Transl Med* 2017;**5**:302.
26. Cheng H, Podolsky DJ, Fisher DM, et al. Teaching palatoplasty using a high-fidelity cleft palate simulator. *Plast Reconstr Surg* 2018;**141**:91e–98ee.

27. Odobescu A, Dawson D, Goodwin I, Harris PG, BouMerhi J, Danino MA. High-fidelity microsurgical simulation: the Thiel cadaveric nerve model and evaluation instrument. *Plast Surg (Oakv)* 2019;**27**:289–96.
28. Kantar RS, Alfonso AR, Ramly EP, et al. Knowledge and skills acquisition by plastic surgery residents through digital simulation training: a prospective, randomized, blinded trial. *Plast Reconstr Surg* 2020;**145**:184e–192ee.
29. Bari AS, Woon CY, Pridgen B, Chang J. Overcoming the learning curve: a curriculum-based model for teaching Zone II flexor tendon repairs. *Plast Reconstr Surg* 2012;**130**:381–8.
30. Saldanha FYL, Loan GJ, Calabrese CE, Sideridis GD, Weinstock PH, Rogers-Vizena CR. Incorporating cleft lip simulation into a "bootcamp-style" curriculum. *Ann Plast Surg* 2021;**86**:210–6.
31. Papavasiliou T, Chatzimichail S, Chan JCY, Bain CJ, Uppal L. A standardized hand fracture fixation training framework using novel 3D printed ex vivo hand models: our experience as a unit. *Plast Reconstr Surg Glob Open* 2021;**9**:e3406.
32. Papavasiliou T, Nicholas R, Cooper L, et al. Utilisation of a 3D printed ex vivo flexor tendon model to improve surgical training. *J Plast Reconstr Aesthet Surg* 2022;**75**:1255–60.
33. Wang C, Chen L, Mu D, Xin M, Luan J. A low-cost simulator for training in endoscopic-assisted transaxillary dual-plane breast augmentation. *Ann Plast Surg* 2017;**79**:525–8.
34. Bauer F, Koerdt S, Hölzle F, Mitchell DA, Wolff KD. Eight free flaps in 24 h: a training concept for postgraduate teaching of how to raise microvascular free flaps. *Br J Oral Maxillofac Surg* 2016;**54**:35–9.
35. Creighton FX, Feng AL, Goyal N, Emerick K, Deschler D. Chicken thigh microvascular training model improves resident surgical skills. *Laryngoscope Invest Otolaryngol* 2017;**2**:471–4.
36. Fuller JC, Justicz NS, Kim J, Cheney M, Castrillon R, Hadlock T. A facial plastic and reconstructive surgery training module using surgical simulation for capacity building. *J Surg Educ* 2019;**76**:274–80.
37. Pafitanis G, Cooper L, Hadjiandreou M, Ghanem A, Myers S. Microvascular anastomotic coupler application learning curve: a curriculum supporting further deliberate practice in ex-vivo simulation models. *J Plast Reconstr Aesthet Surg* 2019;**72**:203–10.
38. Schoeff S, Hernandez B, Robinson DJ, Jameson MJ, Shonka Jr. DC. Microvascular anastomosis simulation using a chicken thigh model: interval versus massed training. *Laryngoscope* 2017;**127**:2490–4.
39. Niitsu H, Hirabayashi N, Yoshimitsu M, et al. Using the objective structured assessment of technical skills (OSATS) global rating scale to evaluate the skills of surgical trainees in the operating room. *Surg Today* 2013;**43**:271–5.
40. Parker RK, Oloo M, Mogambi FC, White RE, Parker AS. Operative self-confidence, hesitation, and ability assessment of surgical trainees in rural Kenya. *J Surg Res* 2021;**258**:137–44.
41. Wiggins-Dohlvik K, Stewart RM, Babbitt RJ, Gelfond J, Zarzabal LA, Willis RE. Surgeons' performance during critical situations: competence, confidence, and composure. *Am J Surg* 2009;**198**:817–23.
42. Elfenbein DM. Confidence crisis among general surgery residents: a systematic review and qualitative discourse analysis. *JAMA Surg* 2016;**151**:1166–75.